

# Sub-threshold Characteristics of N-channel MOSFET



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# Aim of the Experiment

In this experiment, following tasks need to be performed:

- ❶ Obtain the sub-threshold  $I_d$  vs  $V_{gs}$  characteristics of NMOS. This is to be performed in two parts -
  - A) Perform the experiment using an OP-AMP having relatively high noise figure and plot the  $I_d$  vs  $V_{gs}$  characteristics.
  - B) Perform the experiment using an OP-AMP with low noise figure and plot  $I_d$  vs  $V_{gs}$  characteristics and compare the plots obtained.
- ❷ Check the continuity of  $I_d$  vs  $V_{gs}$  characteristics beyond the sub-threshold region.
- ❸ Employing the NMOS to be used as a Common Source Amplifier in sub-threshold region.

# Background Theory

An N-channel MOSFET is Field Effect Transistor which has 4 terminals i.e 1) Drain 2) Gate 3) Source 4) Body. The body terminal of NMOS is connected to lowest voltage possible in the circuit i.e ground. For an NMOS to be 'ON', the applied Gate to Source Voltage must be greater than Threshold Voltage  $V_{th}$  else it was assumed to be 'OFF', i.e.  $V_{gs} \geq V_{th}$ .

In true sense however the MOSFET is not 'OFF' but conducts some small amount of current. When the Gate to Source Voltage is less than threshold voltage, few electrons are still present inside the channel. When Drain Voltage is applied, there exists a depletion region near the drain junction and even lower concentration of minority carriers.

Thus electrons near the Source Junction begins to move towards Drain Junction primarily due to mechanism of Diffusion leading to Sub Threshold Current. The equation of Drain current in Sub Threshold region is given by the following equation.

$$I_d = I_o e^{\left(\frac{V_{gs} - V_{th}}{\eta V_t}\right)} \left(1 - e^{\frac{-V_{ds}}{\eta V_t}}\right)$$

where,

$$I_o = \mu_n C_{ox} \left(\frac{W}{L}\right) V_t^2 (\eta - 1)$$



# Background Theory (Continuation)

- $I_d$  = Drain Current,
- $V_{gs}$  = Gate to Source Voltage,
- $V_{th}$  = Threshold Voltage of NMOS,
- $V_t$  = Thermal Voltage,
- $V_{ds}$  = Drain to Source Voltage,
- $\mu_n$  = Mobility of electron,
- $C_{ox}$  = Gate Oxide Capacitance per unit area,
- $\left(\frac{W}{L}\right)$  = Aspect Ratio of NMOS,
- $\eta = 1 + \left(\frac{C_d}{C_{ox}}\right)$  where  $C_d$  is Depletion Capacitance. It is called as Sub Threshold Slope Factor.

# Approach for designing the experiment

- As seen from equation above, equation for drain current in Sub Threshold region is a very complex equation. Hence before proceeding towards the design part of equation we should try to eliminate at-least one variable (preferably  $V_{ds}$  because we have to plot  $I_d$  vs  $V_{gs}$  characteristics) to make it a function of one variable.
- We start with some assumption of the value of  $\eta$ , say 10. Value of  $\eta V_t = 0.256$  at room temperature. If  $V_{ds} = 2V$ , the exponential term containing  $V_{ds}$  diminishes to zero. Thus the overall term containing  $V_{ds}$  becomes 1. This makes the problem simple. Thus we have,

$$I_d \approx I_o e^{\left(\frac{V_{gs} - V_{th}}{\eta V_t}\right)}$$

- VERY IMP NOTE:** We have assumed the value of  $\eta$  to be 10. After performing the experiment we will back calculate the value of  $\eta$  from some graph. If the value is less than 10, we are safe and the assumption of the exponential term containing  $V_{ds}$  diminishing to zero still holds. If  $\eta$  is greater than 10, we need to pick up another value of  $V_{ds}$  and perform the experiment again.



# Methodology of designing the experiment

- The D.M.M. available in the laboratory/Personal D.M.M. can measure currents accurately to a value of around  $100\ \mu\text{A}$ . However, the Sub Threshold  $I_d$  current which you are about to measure is of order of few hundreds of  $\text{nA}$ .
- Hence we should employ a different methodology for measuring these small currents, perhaps an indirect way of measuring current (i.e measure some Voltage,  $V$  in the order of few volts and thus obtain  $I_d$  from the measured  $V$ ).
- This is done using an OP-AMP and Resistor in Negative Feedback fashion. The Low Noise figure of OP-AMP and High Precision of resistors are very important as the current we are about to measure are small in magnitude. Having an OP-AMP with high noise figure can cause it to be driven into Saturation easily.

# Components Necessary

- CD4007 NMOS I.C.
- TL071 OP-AMP
- TLV9161 OP-AMP
- Keithley Power Supply
- 10 M  $\Omega$  and 1 M  $\Omega$  resistor - 2 quantities
- 100  $\Omega$  Potentiometer
- 3 D.M.M. ( 1 from Lab and 2 from Self )
- Connecting Wires

# Pin Diagram of CD4007

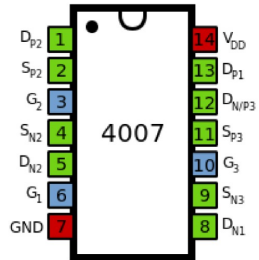
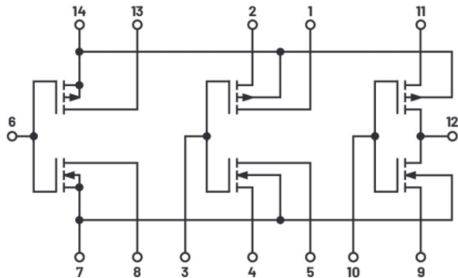


Figure: CD4007 Pinout



# Pin Diagram of TL071 and TLV9161

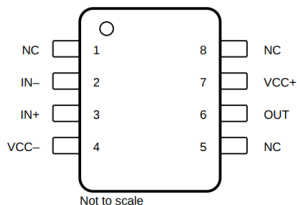


Figure: TL071 Pinout

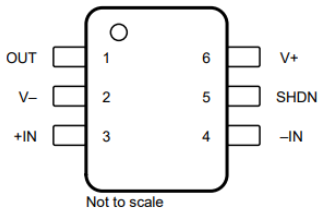


Figure: TLV9161 Pinout in SOT-23 package

# Pin Diagram and functions of TLV9161

| PIN  |     | I/O | DESCRIPTION                                                  |
|------|-----|-----|--------------------------------------------------------------|
| NAME | NO. |     |                                                              |
| +IN  | 3   | I   | Noninverting input                                           |
| −IN  | 4   | I   | Inverting input                                              |
| OUT  | 1   | O   | Output                                                       |
| SHDN | 5   | I   | Shutdown: low = amplifier enabled, high = amplifier disabled |
| V+   | 6   | —   | Positive (highest) power supply                              |
| V−   | 2   | —   | Negative (lowest) power supply                               |

Figure: Pin Functions of TLV9161

# Part - 1: Transfer Characteristics of NMOS (Part A)

Measure  $I_d$  vs  $V_{gs}$  using High Noise Figure OP-AMP (TL-071) in Sub Threshold Region. Follow the steps as below:

- 1 Connect the circuit as shown in the figure below. Follow the pin diagram given in subsection **2.5** for connections. Note that  $V_{cc+}$  should be connected to  $+8V$  and  $V_{cc-}$  should be connected to  $-8V$ .
- 2 Connect the Laboratory D.M.M. such that it measures  $V_{gs}$ .
- 3 Connect the D.M.M. you have purchased to measure  $V_{inv}$  and  $V_{out}$  of OP-AMP.
- 4 If you have a look at the data sheet of CD4007, the threshold voltage of NMOS is somewhere between  $0.4V$  to  $1V$ . Vary  $V_{gs}$  from  $0V$  to value close to the point at what the D.M.M. measuring  $V_{inv}$  is showing. The steps should be of  $0.02V$ . Also measure  $V_{out}$ .  
(However you might not be able to vary  $V_{gs}$  to that point and it will be clarified further why).

# Sub-threshold Measurement using TL-071

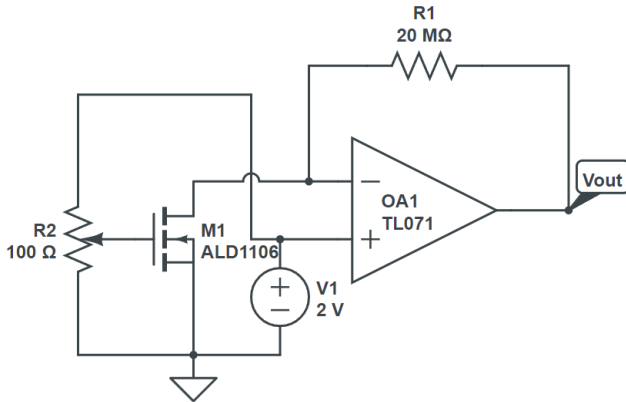


Figure: Circuit Diagram for Sub-threshold Measurement using TL-071

# Part - 1: Transfer Characteristics of NMOS (Part A) - Clarification

## Clarification on Circuit Operation-

- The circuit uses an OP-AMP in negative feedback. The Virtual Short ensures that  $V_{ds} = V_{inv} = 2V$  (which is equal to  $V_{non-inv}$ ).
- The current  $I_d$  which is set up in the circuit is given by the equations as mentioned in the **Approach**. This same current  $I_d$  flows through  $20\text{ M}\Omega$  resistor because the OP-AMP draws no current.
- Equation governing the flow of current through resistor is that of simple Ohm's Law. Hence,

$$I_d = \left( \frac{V_{out} - V_{inv}}{20} \right) \mu A$$

You can convert this to  $nA$  as well. Plot  $I_d$  vs  $V_{gs}$  characteristics.

# Part - 1: Transfer Characteristics of NMOS (Part A) - Points to Remember

## Points to remember-

- Use the Keithley Power Supply for taking the readings.
- It is very important to have a look at values of  $V_{inv}$  and  $V_{out}$ .
- We know that as we increase the value of  $V_{gs}$ , the current  $I_d$  increases. However till the point the Virtual Short is maintained,  $V_{inv}$  will be maintained at potential equal to  $V_{non-inv}$ . This means that as  $I_d$  increases,  $V_{out}$  also increases. It might so happen that at some  $V_{gs}$ , the OP-AMP Voltage  $V_{out}$  saturates. Hence it is important to keep an eye at the value of  $V_{inv}$ . Once the OP-AMP saturates, the Virtual Short fails and hence the claims we made above including Ohm's Law fails. Hence we should stop taking readings beyond that point.

# Continue.....

- Once you see the  $V_{inv}$  ( =  $V_{non-inv}$  ) changing from fixed value or  $V_{out}$  saturating (whichever happens first), one should stop taking the readings.
- Do not change the range settings on D.M.M. while measuring the Voltage values.
- When you start changing the value of Voltage  $V_{gs}$  from 0V to some value  $V_{01}$ , it might so happen that  $V_{out}$  might not change for these range of values of  $V_{gs}$ . This happens because the current  $I_d$  flowing might not be sufficient enough to produce measurable change (measurable change on D.M.M.) in  $V_{out}$ . Once you start seeing some changes in the values of  $V_{out}$ , it is at this point one should start taking the readings. The corresponding  $V_{gs}$  value of should be considered as the first reading and the values of  $V_{gs}$  prior to that must be discarded.
- Make sure to connect the Source and Body terminals of CD4007 NMOS to Ground.

# Part - 1: Transfer Characteristics of NMOS (Part B)

Measure  $I_d$  vs  $V_{gs}$  using Low Noise Figure OP-AMP (TLV9161) in Sub-threshold Region. Repeat same set of steps as mentioned in Part A. Follow the pin diagram given in subsection **2.6** for connections. Note that  $V_+$  should be connected to  $+8V$  and  $V_-$  should be connected to  $-8V$ . Also connect the pin SHDN to  $-8V$  as it enables the OP-AMP. The circuit diagram shown in the following figure.

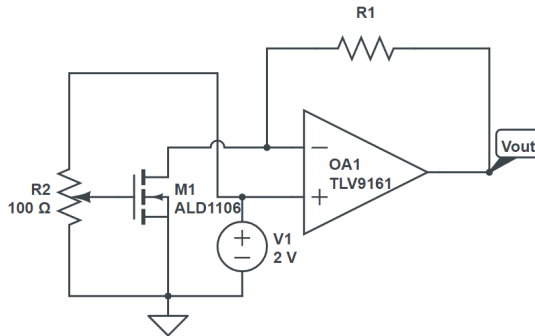


Figure: Circuit Diagram for Sub-threshold Current Measurement



# Continue....

- 1 Initially take  $R_1 = 10M\Omega$ .
- 2 Compute the values of  $I_d$  by changing the value of  $V_{gs}$  in a fashion similar to what we did in PART A. Stop taking the readings when either the OP-AMP saturates or the  $V_{inv}$  starts to change. Make sure not to change the value of  $V_{gs}$  beyond this point.
- 3 Now carefully remove the  $10 M\Omega$  resistance without disturbing the circuit and insert  $R_1 = 1 M\Omega$  in place of  $10 M\Omega$  resistor. Start by taking the values of  $I_d$  from the point where you stopped changing the value of  $V_{gs}$  for  $R = 10 M\Omega$  resistor (This is for maintaining continuity in the readings). Now increase the value of  $V_{gs}$  and keep on taking the readings again till the OP-AMP saturates or  $V_{Inverting}$  changes. Note the value of  $V_{gs}$  where you have stopped taking the readings.
- 4 Concatenate the values of  $I_d$  received from  $10 M\Omega$  and  $1 M\Omega$  resistors and plot the resultant  $I_d$  vs  $V_{gs}$  graph.

# Part - 1: Transfer Characteristics of NMOS (Part B) - Conclusions

## Conclusions you need to draw from the two sets of readings-

- 1 Plot the  $I_d$  vs  $V_{gs}$  Sub Threshold Characteristics obtained from both the OP-AMPs on a single graph and compare both the characteristics. The OP-AMP with high noise figure tends to saturate early and gives you less accurate readings than one with low noise figure. This is especially important when you are designing circuits employing semiconductor devices having bias currents and voltages of very small orders. Hence proper selection of components becomes crucial at such times.
- 2 For Part B readings, one should also plot the  $\log_{10} I_d$  vs  $V_{gs}$  and measure the Sub threshold Swing. Remember we have assumed the value of  $\eta$  and we need to cross verify if the assumed value holds or not. For this we will calculate the value of Sub Threshold Swing,  $S$  which is equal to -

$$S = \frac{dV_{gs}}{d(\log_{10} I_d)}$$

- 1 The value of  $S$  is obtained by taking the slope of the graph  $\log_{10} I_d$  vs  $V_{gs}$  and taking inverse of it. Make sure you express the value of  $S$  in  $\frac{mV}{decade}$ . The slope of  $\log_{10} I_d$  vs  $V_{gs}$  is also known as Sub Threshold Slope.
- 2 For a long channel transistor, the value of  $S$  is around  $60 \frac{mV}{decade}$ . This assumes that the value of  $\eta$  is 1. For a FET such as CD4007 the value of  $S$  is equal to,

$$S = 60\eta \frac{mV}{decade}$$

# Part - 1: Transfer Characteristics of NMOS (Part B) - Conclusions (Continuation)

- 5 From this we can calculate the value of  $\eta$ . If the value of  $\eta \geq 10$ , we need to change the value of  $V_{ds}$  and repeat the whole process again. Else we are good with the readings.
- 6 The graph of  $\log_{10} I_d$  vs  $V_{gs}$  cuts the Y-axis at some point. Let us call that point  $I_{off}$ . It is called as Off Current of NMOS. This becomes an important parameter in circuit design. Basically it tells us what would be the value of current  $I_d$  flowing through the MOSFET even if the value of  $V_{gs} = 0$ . This gives an idea about Static Power dissipation in the circuits which uses MOSFETs. State the value of  $I_{off}$  obtained. Also calculate the power dissipated in the MOSFET under such conditions. Any value greater than few  $nW$  is dangerously high!

## Part - 2: Graph continuity

- 1 Remember for **Part B** of the readings you were asked to stop taking readings for some  $V_{gs}$  for the case of  $R1 = 1M\Omega$ . We will be making measurements beyond this value now. This  $V_{gs}$  would be somewhere around  $0.7V - 0.8V$ . Increase the value of  $V_{gs}$  beyond this point using the potentiometer.
- 2 Also note that by this point  $V_{gs}$  value is sufficiently high such that we do not need the circuit in Part - 1 of the measurements. We can directly measure  $I_d$  because currents would be in order of few  $\mu A$ .
- 3 Make the connections as shown in the figure below-
- 4 Increase the value of  $V_{gs}$  and measure corresponding value of  $I_d$ . The step size in which you will increase  $V_{gs}$  should be of  $0.1V$ . Stop taking readings when  $V_{gs}$  is around  $1.9V$ .

# Testing Continuity Characteristics

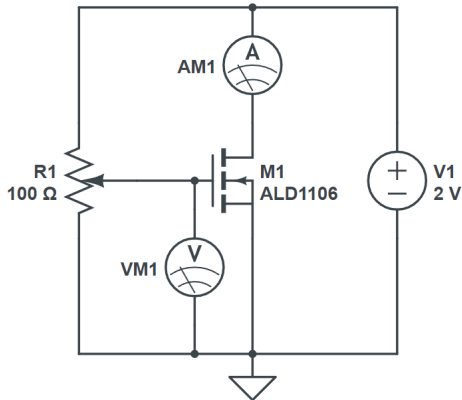


Figure: Circuit Diagram for testing continuity of Characteristics

## Part - 2: Graph continuity - Conclusions

Conclusions to be drawn from the readings -

- 1 Carefully note that once the  $V_{gs}$  value crosses  $1V$  we are somewhat sure that NMOS is not in Sub-threshold region. It is in some other region. Comment down what region NMOS is in in this case.
- 2 In the Part-B of the Part-1 of the experiment you measured  $I_d$  vs  $V_{gs}$  for  $R = 1M\Omega$  and  $R = 10 M\Omega$  and concatenated the same in one single graph. Now you need to concatenate the values of Part-B and the readings obtained in Part-2 of experiment and plot  $I_d$  vs  $V_{gs}$  for entire range of values of  $V_{gs}$  on single graph and check if continuity in readings exist. Also plot  $\log_{10} I_d$  vs  $V_{gs}$  and explain the shape of this graph.

# Part - 3: Common Source Amplifier using MOSFET biased in Sub-Threshold

- You would have heard of Common Source Amplifier which makes use of MOSFET in Saturation Region. One can manage to get a decent gain of roughly around 15 to 20 using discrete transistors and components. However, as our technology nodes become smaller, with it comes the challenge of using smaller and smaller voltages to ensure our transistors are operating in Saturation. This leaves with very little headroom for our transistors to manage their output swings to stay in Saturation region and also strains power dissipation that can happen in the circuit. So how about using transistor in Sub-Threshold region?
- We already know that Sub-Threshold regions use smaller Voltages. Coupled to it the currents are in order of few  $nA$ . This can help us design an amplifier. Even though the gain may not be great, we can always cascade amplifiers and get desired gain.





# Part - 3: Common Source Amplifier - Steps

Steps to follow -

- 1 Connect the circuit as shown in the figure. Set the  $V_{gs}$  value by making D.C. Offset on function generator = 400 mV. Do not provide sine wave as of now. Also make sure not to connect the  $V_{out}$  to D.S.O.
- 2 Connect a D.M.M. to the drain node and vary V3 till the D.M.M. reads 2V. This sets the  $V_{ds}$  to 2V. Remove the D.M.M.
- 3 For Input Small Signal settings, set the function generator so that it applies a sine wave of frequency 50Hz, 50mV Peak to Peak. Let the D.C. Offset be 400 mV. The D.C. offset of 400 mV will be superimposed on the sine wave.
- 4 Now connect the  $V_{out}$  node to D.S.O. and observe the Peak to Peak Voltage of the  $V_{out}$ . Note down the Value.
- 5 Gain of Amplifier =  $\frac{V_{out(p-p)}}{V_{in(p-p)}}$
- 6 Report the gain.

# Common Source Amplifier

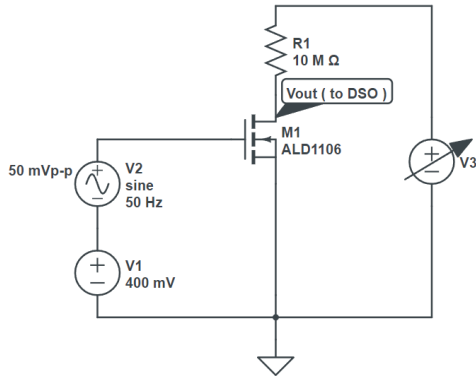


Figure: Common Source Amplifier Design

# Part - 3: Common Source Amplifier - Conclusions

Conclusions to be drawn from the circuit -

- ① Gain of a Basic Common Source Amplifier in Sub-threshold Region might be poor compared to Saturation Region of Operation. But it is respectable enough to allow it to be used as an Amplifier.
- ② Note that the currents involved in Sub-threshold Region of Operation are of the order of few  $nA$ . Hence the current driving capability of the amplifiers are pretty much limited unlike the Saturation region. Which is why we cannot use them in applications which needs high current drives.